Analysis part3

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Categorical variables

For categorical variables, common summary statistics include:

- **Frequencies** (counts): number of observations in each category
- **Proportions** (percentages): share of each category relative to the total.
- **Mode**: Most frequent category.
- **Contingency tables** (cross-tabulations): show the relationship between two or more categorical variables.

Chi-square test

statistical test for independence between categorical variables

Types of Chi squared tests

- **Chi-square test of independence**: tests if two variables are related (e.g., gender and preference for a product)
- **Chi-square goodness-of-fit test**: tests if the observed distribution fits a specific expected distribution

Chi squared for independence

How it works: It compares the observed counts in a contingency table to the expected counts (if there were no association). The formula is:

Assumptions:

- Categories are mutually exclusive
- Expected frequencies should generally be ≥ 5
- Observations are independent

$$Chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

O = observed count

E = expected count

```r
library(readxl)
melanoma <- read_excel("melanoma_dataset.xlsx")
melanoma$status <- as.factor(melanoma$status)
table(melanoma$sex, melanoma$status)
```

```
##
##      1  2  3
##   0 28 91  7
##   1 29 43  7
```

```r
chisq.test(table(melanoma$sex, melanoma$status))
```

```
##
##     Pearson's Chi-squared test
##
## data:  table(melanoma$sex, melanoma$status)
## X-squared = 6.793, df = 2, p-value = 0.03349
```

Numeric variables (continue)

The correlation coefficient measures the strength and direction of a linear relationship between two continuous variables

Values range from –1 to +1:

- **+1**: perfect positive linear correlation
- **–1**: perfect negative linear correlation
- **0**: no linear correlation

The most common is Pearson's correlation coefficient (r) - assumption of parametric data.

Pearson's correlation

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$$

Example

```r
# data
x <- c(1.5, 4.5, 6.3, 8.4, 10)
y <- c(1, 2, 6, 3, 9)

# Pearson correlation
cor(x, y, method = "pearson")
```

```
## [1] 0.7941187
```

Testing Pearson's correlation

```r
# Correlation test
cor.test(x, y, method = "pearson")
```

```
##
##     Pearson's product-moment correlation
##
## data:  x and y
## t = 2.2631, df = 3, p-value = 0.1086
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.2944386  0.9857469
## sample estimates:
##       cor
## 0.7941187
```

What if data is non parametric? Spearman's correlation

- If not non-parametric it does not assume normal distribution
- Values range from –1 to +1, just like Pearson's

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

It captures any consistent increasing or decreasing trend, not necessarily linear

```
# Spearman correlation
cor(x, y, method = "spearman")
```

```
## [1] 0.9
```

Testing Spearman's correlation

```r
# Correlation test
cor.test(x, y, method = "spearman")
```

```
##
##     Spearman's rank correlation rho
##
## data:  x and y
## S = 2, p-value = 0.08333
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.9
```

Spearman over Pearson

No Assumption of Linearity

Spearman detects monotonic relationships (increasing or decreasing), not just linear ones. Pearson only measures linear relationships.

Robust to Outliers

Spearman uses ranks, so extreme values have less influence. Pearson is sensitive to outliers, which can distort results.

Works with Ordinal Data

Spearman can be used when variables are ordinal (ranked categories). Pearson requires interval or ratio (continuous) data.

Better with heteroscedasticity

Spearman is more robust when the spread of data varies across the range. Pearson assumes homoscedasticity (equal variance).

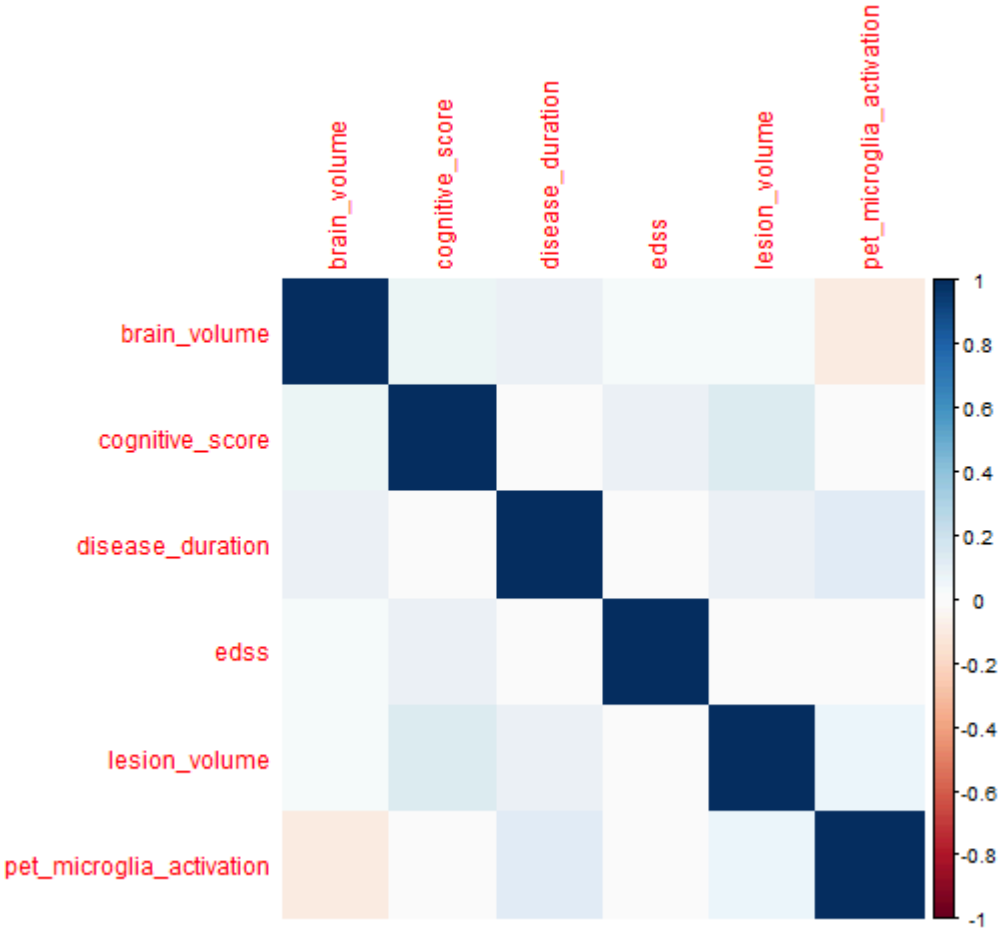
Corplot

```
library(corrplot)
```

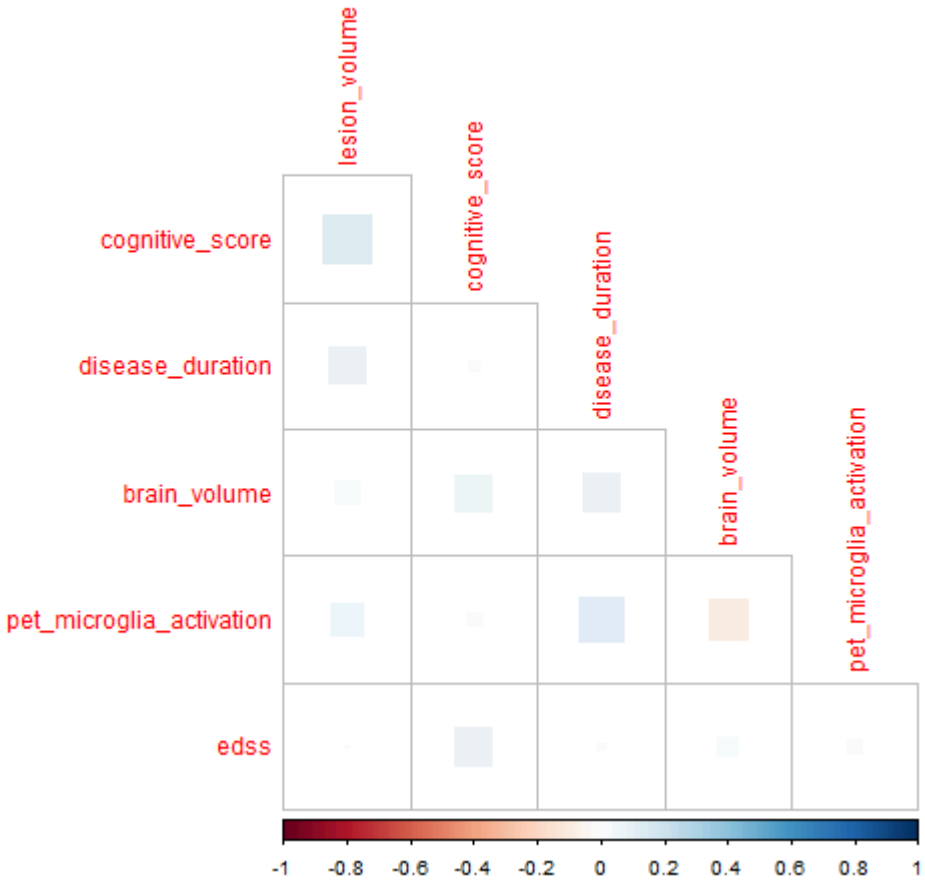
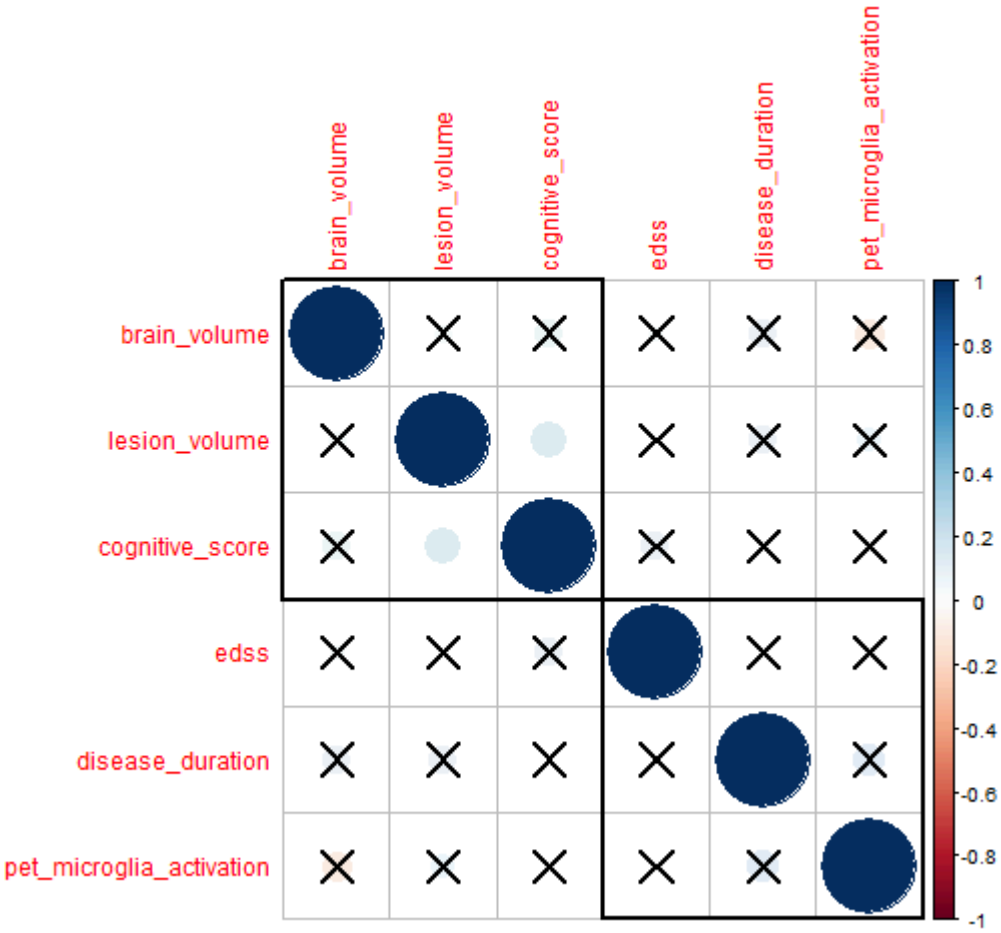
```
## corrplot 0.95 loaded
```

```
neurological_examination <- read_excel("neurological_examination.xlsx
neuro.num <- neurological_examination[, 3:8]
M = cor(neuro.num)
```

```r
corrplot(M, method = 'color', order = 'alphabet')
```



```
corrplot(M, method = 'square', order = 'FPC', type = 'lower', diag =
```



```
testRes = cor.mtest(neuro.num, conf.level = 0.95)

## specialized the insignificant value according to the significant l
corrplot(M, p.mat = testRes$p, sig.level = 0.10, order = 'hclust', ac
```